

SDET TRAINING PORTFOLIO

Comprehensive UI Automation Framework Specification

Technical Documentation of Consolidated Component Testing and Advanced User Interactions

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Package Structure	Sourav_Wipro.Selenium.Project6
Target Sandbox	testautomationpractice.blogspot.com
Core Dependencies	WebDriverManager v5.x / Selenium WebDriver 4.x

1. Architectural Feature Mapping Matrix

The operational specifications of the automation suit are mapped systematically across eleven mandatory UI concepts to trace script actions back to explicit framework capabilities:

#	Core Automation Concept	Implementation Strategy	Selenium API Method
1	Text Fields	Target attributes by placeholder or ID; pass keyboard strings.	<code>sendKeys()</code>
2	Buttons	Isolate action nodes via class name or absolute structural XPath lines.	<code>click()</code>
3	Radio Buttons	Select gender metrics by unique ID matching targets.	<code>click()</code>
4	List Boxes	Extract multiple continuous element tags using shared descriptors.	<code>findElements(By.tagName())</code>
5	Dropdown Buttons	Wrap elements inside the explicit Select structure; extract targets by exact visible names.	<code>Select.selectByVisibleText()</code>
6	Image Handling	Interact directly with the file type node; push absolute text files separated by newlines.	<code>sendKeys(path1 + "\n" + path2)</code>
7	Check Boxes	Select day choices by explicit ID matching properties.	<code>click()</code>
8	Scroll Page / Execution	Inject JavaScript scripts to cast values into custom target fields or control viewports.	<code>JavascriptExecutor.executeScript()</code>
9	Screen Outputs / Attributes	Pull text nodes or raw form input values down into consoles.	<code>getText(), getAttribute("value")</code>
10	Actions Pointer Class	Build mouse actions (Double click, Context click, Hover, Drag-and-Drop, Offset Slider movement).	<code>Actions.perform()</code> chains
11	Modal Alerts	Shift engine tracking states to capture, read, and accept or reject alerts.	<code>driver.switchTo().alert()</code>

2. Technical Step-by-Step Implementation

The breakdown below details the execution steps of the consolidated test runner class:

Step 1 Form Environment Initialization

Initialize ChromeDriver with WebDriverManager binaries, launch the screen maximize routine, fetch the target sandbox, and output the tracking main header node to the text stream.

```
WebDriverManager.chromedriver().setup();
WebDriver driver = new ChromeDriver();
driver.manage().window().maximize();
driver.get("https://testautomationpractice.blogspot.com/");
WebElement wb = driver.findElement(By.xpath("//h1[contains(text(), 'Automation Testing Practice')]"));
System.out.println(wb.getText());
```

Step 2 Standard Form Field Ingestion

Populate text boxes, textareas, radial parameters, and checkboxes using specific locators.

```
driver.findElement(By.xpath("//input[@placeholder='Enter Name']")).sendKeys("Sourav Singh");
driver.findElement(By.id("email")).sendKeys("heaveneater@gmail.com");
driver.findElement(By.id("phone")).sendKeys("1234567890");
driver.findElement(By.id("textarea")).sendKeys("Jharsugda, Odisha");
driver.findElement(By.id("male")).click();
driver.findElement(By.id("sunday")).click();
```

Step 3 Dropdown and List Box Interactivity

Wrap standard select components inside the structured Select abstraction class to isolate items by visible text properties.

```
Select country = new Select(driver.findElement(By.xpath("//select[contains(@id, 'country')]")));
country.selectByVisibleText("India");

Select colors = new Select(driver.findElement(By.xpath("//select[contains(@id, 'colors')]")));
colors.selectByVisibleText("White");

Select animals = new Select(driver.findElement(By.id("animals")));
animals.selectByVisibleText("Rabbit");
```

Step 4 Dynamic Date Field Handling via Native and JavaScript Pipelines

Populate target calendar inputs using native keyboard streams or direct DOM value injections with JavaScript execution hooks.

```
driver.findElement(By.id("datepicker")).click();
driver.findElement(By.id("datepicker")).sendKeys("03/25/2026");
driver.findElement(By.xpath("//label[contains(text(),'Date Picker 3')]")).click();

JavascriptExecutor js = (JavascriptExecutor) driver;
WebElement date = driver.findElement(By.id("txtDate"));
js.executeScript("arguments[0].value='09/09/2021';", date);

driver.findElement(By.id("start-date")).sendKeys("12-05-2004");
driver.findElement(By.id("end-date")).sendKeys("12-05-2005");
driver.findElement(By.xpath("//button[@class='submit-btn']")).click();
```

Step 5 Multi-Context File and Media Attachment Stream

Bypass native window prompts by pushing absolute storage file reference strings, delimited with newline characters, directly into the underlying input fields.

```
String singleFilePath = new File("C:\\Users\\SOURAV SINGH\\OneDrive\\Desktop\\Wipro
Assignments\\1776959736860-Assignment SDET - Day 1.pdf").getAbsolutePath();
String file1Path = "C:\\Users\\SOURAV SINGH\\OneDrive\\Desktop\\Wipro Assignments\\
1778330467761-Assignment APRIL SDET - Day 15.pdf";
String file2Path = "C:\\Users\\SOURAV SINGH\\OneDrive\\Pictures\\images.jpg";

WebElement singleInput = driver.findElement(By.id("singleFileInput"));
singleInput.sendKeys(singleFilePath);
driver.findElement(By.xpath("//form[@id='singleFileForm']/button")).click();

WebElement multipleInput = driver.findElement(By.id("multipleFilesInput"));
multipleInput.sendKeys(file1Path + "\n" + file2Path);
driver.findElement(By.xpath("//form[@id='multipleFilesForm']/button")).click();
```

Step 6 Advanced User Gesture Control via Actions API

Instantiate a specialized Actions builder to simulate composite input sequences, including double-clicks, right-clicks, hover effects, drag-and-drop operations, and explicit pixel offsets for sliders.

```
Actions act = new Actions(driver);

WebElement copyBtn = driver.findElement(By.xpath("//button[text()='Copy Text']"));
act.doubleClick(copyBtn).perform();
act.contextClick(copyBtn).perform();

WebElement pointMe = driver.findElement(By.xpath("//button[text()='Point Me']"));
act.moveToElement(pointMe).perform();

WebElement source = driver.findElement(By.id("draggable"));
WebElement target = driver.findElement(By.id("droppable"));
act.dragAndDrop(source, target).perform();

WebElement slider = driver.findElement(By.xpath("//span[@class='ui-slider-handle ui-corner-all ui-state-default']"));
act.dragAndDropBy(slider, 100, 0).perform();
```

Step 7 Dynamic Links Assessment and Value Captures

Collect all interactive references using global tag scanning routines and log field state properties before starting alert checks.

```
List<WebElement> links = driver.findElements(By.tagName("a"));
System.out.println("Total Links : " + links.size());

String text = driver.findElement(By.id("name")).getAttribute("value");
System.out.println(text);
```

Step 8 Tri-Modal Native Alert Handlers

Trigger and manage alternative alert interactions by shifting the target context to handle Information, Confirmation, and Prompt modals sequentially.

```
// Simple Information Alert Handling
driver.findElement(By.id("alertBtn")).click();
Alert a = driver.switchTo().alert();
System.out.println(a.getText());
a.accept();

// Two-Option Confirmation Modal Dismissal
driver.findElement(By.id("confirmBtn")).click();
Alert b = driver.switchTo().alert();
System.out.println(b.getText());
b.dismiss();

// Input Prompt Modal Injection and Acceptance
driver.findElement(By.id("promptBtn")).click();
Alert c = driver.switchTo().alert();
System.out.println(c.getText());
c.sendKeys("Sourav Singh");
c.accept();
```

Design Refinement Recommendation: Replace hardcoded `Thread.sleep()` invocations with dedicated, condition-driven `WebDriverWait` or implicit timeout limits to build non-blocking synchronization layers.

3. Clean Monolithic Execution Engine

The integrated production runner source code below matches your exact code profile, organized without inner narrative blocks to maintain professional readability:

```
package Sourav_Wipro.Selenium;

import java.io.File;
import java.util.List;
import org.openqa.selenium.Alert;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.interactions.Actions;
import org.openqa.selenium.support.ui.Select;
import org.openqa.selenium.JavascriptExecutor;
import io.github.bonigarcia.wdm.WebDriverManager;

public class Project6 {
```

```

public static void main(String[] args) throws Exception {

    WebDriverManager.chromedriver().setup();
    WebDriver driver = new ChromeDriver();
    driver.manage().window().maximize();
    driver.get("https://testautomationpractice.blogspot.com/");

    WebElement wb = driver.findElement(By.xpath("//h1[contains(text(),'Automation Testing Practice')]"));
    System.out.println(wb.getText());

    driver.findElement(By.xpath("//input[@placeholder='Enter Name']")).sendKeys("Sourav Singh");
    Thread.sleep(2000);

    driver.findElement(By.id("email")).sendKeys("heaveneater@gmail.com");
    Thread.sleep(2000);

    driver.findElement(By.id("phone")).sendKeys("1234567890");
    Thread.sleep(2000);

    driver.findElement(By.id("textarea")).sendKeys("Jharsugda, Odisha");
    Thread.sleep(2000);

    driver.findElement(By.id("male")).click();
    Thread.sleep(2000);

    driver.findElement(By.id("sunday")).click();
    Thread.sleep(2000);

    Select country = new Select(driver.findElement(By.xpath("//select[contains(@id,'country')]")));
    country.selectByVisibleText("India");
    Thread.sleep(2000);

    Select colors = new Select(driver.findElement(By.xpath("//select[contains(@id,'colors')]")));
    colors.selectByVisibleText("White");
    Thread.sleep(2000);

    Select animals = new Select(driver.findElement(By.id("animals")));
    animals.selectByVisibleText("Rabbit");
    Thread.sleep(2000);

    driver.findElement(By.id("datepicker")).click();
    driver.findElement(By.id("datepicker")).sendKeys("03/25/2026");
    Thread.sleep(2000);
    driver.findElement(By.xpath("//label[contains(text(),'Date Picker 3')]")).click();

    JavascriptExecutor js = (JavascriptExecutor) driver;
    WebElement date = driver.findElement(By.id("txtDate"));
    js.executeScript("arguments[0].value='09/09/2021';", date);
    Thread.sleep(2000);
}

```

```

driver.findElement(By.id("start-date")).sendKeys("12-05-2004");
Thread.sleep(2000);

driver.findElement(By.id("end-date")).sendKeys("12-05-2005");
Thread.sleep(2000);

driver.findElement(By.xpath("//button[@class='submit-btn']")).click();
Thread.sleep(2000);

String singleFilePath = new File("C:\\Users\\SOURAV SINGH\\OneDrive\\Desktop\\Wipro
Assignments\\1776959736860-Assignment SDET - Day 1.pdf").getAbsolutePath();
String file1Path = new File("C:\\Users\\SOURAV SINGH\\OneDrive\\Desktop\\Wipro
Assignments\\1778330467761-Assignment APRIL SDET - Day 15.pdf").getAbsolutePath();
String file2Path = new File("C:\\Users\\SOURAV SINGH\\OneDrive\\Pictures\\
\\images.jpg").getAbsolutePath();

WebElement singleInput = driver.findElement(By.id("singleFileInput"));
singleInput.sendKeys(singleFilePath);

WebElement singleSubmitBtn = driver.findElement(By.xpath("//
form[@id='singleFileForm']/button"));
singleSubmitBtn.click();

WebElement multipleInput = driver.findElement(By.id("multipleFilesInput"));
String multipleFilesPaths = file1Path + "\\n" + file2Path;
multipleInput.sendKeys(multipleFilesPaths);

WebElement multipleSubmitBtn = driver.findElement(By.xpath("//
form[@id='multipleFilesForm']/button"));
multipleSubmitBtn.click();

Actions act = new Actions(driver);

WebElement copyBtn = driver.findElement(By.xpath("//button[text()='Copy Text']"));
act.doubleClick(copyBtn).perform();
System.out.println("Double Click Done");

act.contextClick(copyBtn).perform();
Thread.sleep(2000);
System.out.println("Right Click Done");

WebElement pointMe = driver.findElement(By.xpath("//button[text()='Point Me']"));
act.moveToElement(pointMe).perform();
Thread.sleep(2000);
System.out.println("Mouse Hover Done");

WebElement source = driver.findElement(By.id("draggable"));
WebElement target = driver.findElement(By.id("droppable"));
act.dragAndDrop(source, target).perform();
Thread.sleep(2000);
System.out.println("Drag and Drop Done");

```



```

        WebElement slider = driver.findElement(By.xpath("//span[@class='ui-slider-handle ui-corner-all ui-state-default']"));
        act.dragAndDropBy(slider, 100, 0).perform();
        Thread.sleep(2000);
        System.out.println("Slider Moved");

        List<WebElement> links = driver.findElements(By.tagName("a"));
        Thread.sleep(2000);
        System.out.println("Total Links : " + links.size());

        String text = driver.findElement(By.id("name")).getAttribute("value");
        System.out.println(text);

        driver.findElement(By.id("alertBtn")).click();
        Alert a = driver.switchTo().alert();
        System.out.println(a.getText());
        Thread.sleep(3000);
        a.accept();
        Thread.sleep(2000);

        driver.findElement(By.id("confirmBtn")).click();
        Alert b = driver.switchTo().alert();
        System.out.println(b.getText());
        Thread.sleep(3000);
        b.dismiss();
        Thread.sleep(3000);

        driver.findElement(By.id("promptBtn")).click();
        Alert c = driver.switchTo().alert();
        System.out.println(c.getText());
        c.sendKeys("Sourav Singh");
        Thread.sleep(2000);
        c.accept();
        Thread.sleep(3000);

        driver.quit();
    }
}

```